

FinaleDB: a browser and database of cell-free DNA fragmentation patterns

Haizi Zheng¹, Michelle S Zhu², Yaping Liu^{1,3-5,*}

1. Division of Human Genetics, Cincinnati Children's Hospital Medical Center, Cincinnati, OH 45229, 2. Department of Computer Science, University of Texas at Austin, Austin, TX 78712, 3. Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center, Cincinnati, OH 45229, 4. Department of Pediatrics, University of Cincinnati College of Medicine, Cincinnati, OH 45229, 5. Department of Electrical Engineering and Computing Sciences, University of Cincinnati College of Engineering and Applied Science, Cincinnati, OH 45229.

*To whom correspondence should be addressed.

Supplementary Information

1. Data collection

In the current version, we chose 7 publications as the source of our cfDNA fragmentation data, from which we collected a total of 2579 raw paired-end sequencing datasets, as summarized below in Supplementary Table 1:

Publication	Count of datasets
Adalsteinsson, Viktor A., et al. "Scalable whole-exome sequencing of cell-free DNA reveals high concordance with metastatic tumors." <i>Nature communications</i> 8.1 (2017): 1-13.	1508
Cristiano, Stephen, et al. "Genome-wide cell-free DNA fragmentation in patients with cancer." <i>Nature</i> 570.7761 (2019): 385-389.	540
Jiang, Peiyong, et al. "Lengthening and shortening of plasma DNA in hepatocellular carcinoma patients." <i>Proceedings of the National Academy of Sciences</i> 112.11 (2015): E1317-E1325.	225
Burnham, Philip, et al. "Urinary cell-free DNA is a versatile analyte for monitoring infections of the urinary tract." <i>Nature communications</i> 9.1 (2018): 1-10.	141
Rabinowitz, Tom, et al. "Bayesian-based noninvasive prenatal diagnosis of single-gene disorders." <i>Genome research</i> 29.3 (2019): 428-438	74
Snyder, Matthew W., et al. "Cell-free DNA comprises an in vivo nucleosome footprint that informs its tissues-of-origin." <i>Cell</i> 164.1-2 (2016): 57-68.	60
Sun, Kun, et al. "Orientation-aware plasma cell-free DNA fragmentation analysis in open chromatin regions informs tissue of origin." <i>Genome research</i> 29.3 (2019): 418-427.	31
Total	2579

Supplementary Table 1. Publications from which the cfDNA fragment data are collected.

We downloaded the raw sequence files (FASTQ) from the National Center for Biotechnology Information (NCBI)'s Gene Expression Omnibus (GEO) and the European Bioinformatics Institute (EMBL-EBI)'s European Genome-Phenome Archive (EGA).

The samples cover a wide range of 23 different pathological conditions (Supplementary Table 2).

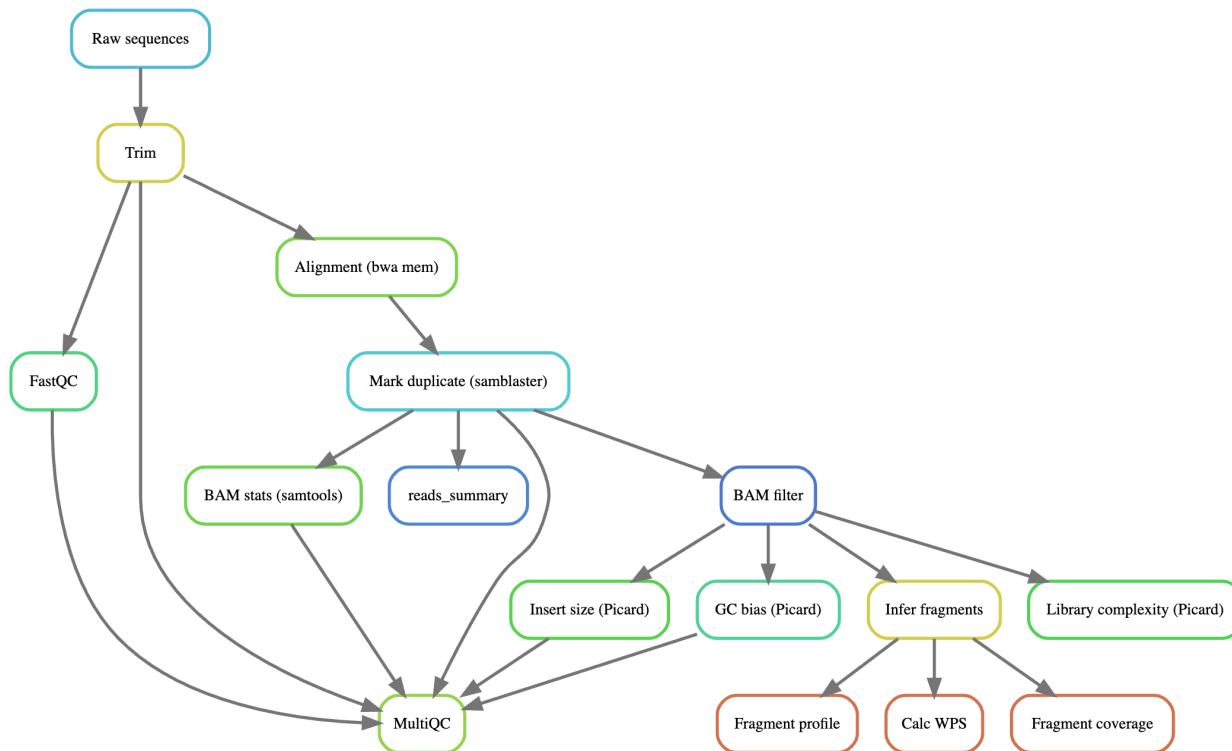
Pathological condition	Count of sequencing records
Breast cancer	1010
Prostate cancer	561
Healthy	298
Liver cancer	93
Lung cancer	88
Hepatitis B	67
Colorectal cancer	48
Pancreatic cancer	43
Cirrhosis	36
Ovarian cancer	30
Gastric cancer	27
Bile duct cancer	25
Liver transplant	13
Systemic lupus erythematosus	4
Inflammatory bowel disease	4
Esophageal cancer	2
Testicular cancer	2
Skin cancer	2
Bladder cancer	2
Uterine cancer	2
Kidney cancer	2
Head and neck cancer	2
Duodenal cancer	1
Grand Total	2362

Supplementary Table 2. Pathological conditions covered by FinaleDB datasets.

2. Data process

We developed a data process pipeline to analyze all the raw sequencing data (FASTQ) files. This strategy allows us to minimize potential batch effects across samples from different studies. The pipeline uses snakemake v5.19 (<https://snakemake.readthedocs.io/en/stable/>) as the workflow management tool. It is open sourced at https://github.com/epifluidlab/finaledb_workflow.

The whole workflow is demonstrated as below in Supplementary Figure 1:



Supplementary Figure 1. Overview of the data process workflow.

2.1. Trim

We used trimmomatic v0.39 (<http://www.usadellab.org/cms/?page=trimmomatic>) to perform quality trim and Illumina adaptor clipping. It also crops all reads to 50bp, in order to reduce potential batch effects.

The arguments we used:

```
CROP:50 ILLUMINACLIP:TruSeq3-PE-2:2:30:10:2:keepBothReads MINLEN:36
```

The trimmomatic adapter file we used: <https://github.com/timflutre/trimmomatic/blob/master/adapters/TruSeq3-PE-2.fa>

2.2. Alignment

We used bwa mem v0.7.17 (<https://github.com/lh3/bwa>) to perform the reads alignment, with the following reference genome assemblies:

- For hg19: ftp://ftp-trace.ncbi.nih.gov/1000genomes/ftp/technical/reference/human_g1k_v37.fasta.gz
- For hg38: ftp://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/GCA_000001405.15_GRCh38_no_alt_analysis_set.fna.gz

2.3. Mark duplicates

We used samblaster v0.1.25 (<https://github.com/GregoryFaust/samblaster/>) to mark PCR duplicates. Compared with other duplicate marking tools such as Picard MarkDuplicates, samblaster performs in a single-pass manner and thus is more effective in certain scenarios.

Entries in the BAM file are marked as duplicates will further be removed in the BAM filter step.

2.4. BAM filter

In order to perform downstream analysis including GC bias, fragment inference, we first filtered the BAM file with the following argument

```
samtools view -f 3 -F 3852
```

This filter rule excluded any reads which:

- Are not properly paired, or
- Are secondary alignments, or
- Failed platform/vendor quality checks, or
- Are PCR or optical duplicate (marked duplicates), or
- Are part of a chimeric alignment (supplementary alignment)

2.5. GC bias

We used Picard Tools v2.22.3 (<https://github.com/broadinstitute/picard/>) to perform the GC bias analysis, with the following arguments:

```
-XX:+UseParallelGC -Dpicard.useLegacyParser=false \  
-Xms512m -Xmx16000m -XX:MaxMetaspaceSize=768m \  
CollectGcBiasMetrics -I {input.bam} -R {input.ref}
```

Here, {input.ref} indicates the reference genome used in the corresponding alignment.

2.6. Library complexity estimation

We used Picard Tools v2.22.3 (<https://github.com/broadinstitute/picard/>) to estimate the library complexity, which is defined as the number of unique fragments in the library, with PCR duplicates excluded. A library with low complexity may indicate high PCR duplicate ratio and may negatively affect the downstream analysis.

We estimated the library complexity with the following arguments:

We used Picard Tools v2.22.3 (<https://github.com/broadinstitute/picard/>) to perform the GC bias analysis, with the following argument:

```
-XX:+UseParallelGC -Dpicard.useLegacyParser=false \  
-Xms512m -Xmx16000m -XX:MaxMetaspaceSize=768m \  
EstimateLibraryComplexity -I {input.bam} -VALIDATION_STRINGENCY SILENT
```

2.7. Fragment inference

From each filtered BAM file, we extracted the outer fragment endpoint coordinates for each pair of aligned reads. The insert between the paired endpoints are recognized as a fragment, and the interval is reckoned as the fragment length.

The fragment data is stored in a tab-separated value (TSV) file, with the same column definition as BedGraph format. Each fragment data file contains five columns (Supplementary Table 3):

Column	Description
1	Chromosome name
2	Start coordinate of the fragment
3	End coordinate of the fragment
4	MAPQ score of the alignment (if the two paired reads have different MAPQ scores, we choose the smaller one)
5	Strand to which the first read is mapped

Supplementary Table 3. The columns in the fragment data file.

We choose the UCSC 0-based half-open convention for the coordinate system.

2.8. Fragment coverage profile

To calculate the fragment coverage, we first ruled out fragments with MAPQ scores less than 30, which either indicate low mapping qualities, or reads not uniquely mapped to the reference genome (when MAPQ equals 0).

We then defined a fragment's coverage as all positions that lie between its two ends, with the endpoints included.

2.9. Fragment length profile

Similar to the fragment coverage calculation, we excluded all fragments with MAP scores less than 30.

The fragment length is defined as the interval between fragment ends, with both endpoints included, that is, the difference between Column 2 and Column 3 in the fragment data file.

2.10. Windowed Protection Score (WPS)

Similar to the fragment coverage calculation, we excluded all fragments with MAP scores less than 30. Moreover, we exclude all fragments that are shorter than 120bp or longer than 180bp.

For a given coordinate in the genome, we first defined a 120bp window symmetric to the said coordinate. We then calculated the WPS which is defined as the number of fragments completely spanning the window, minus the number of fragments with one of their endpoints located within the same window.

2.11. The coverage of fragment starting and ending position

Similar to the fragment coverage calculation, we excluded all fragments with MAP scores less than 30.

We provided a R script in the associated github repository to calculate the coverage of starting and ending position of fragments:

https://github.com/epifluidlab/finaledb_workflow/blob/master/endpoint_coverage.R

More details can be found in the README file: https://github.com/epifluidlab/finaledb_workflow#calculate-the-endpoint-coverage

3. Web portal manual

3.1. Query

To the left of the page is the filter pane, where users can make database queries. In the “Search” text box, users can search by sample names (e.g. MBC_1292), sequencing run ID (e.g. EE86284), or SRA ID (e.g. SRR6708248).

In addition to the search box, users can also search through the facet filter described below,

- Platform: the sequencer used to generate the data
- Assay type: currently there are two assay types in the database: whole genome sequencing (WGS) and whole exome sequencing (WES)
- Min/max # of fragments: specify a range and only return the records with the number of fragments within
- Disease status: return records associated with any particular pathological conditions
- Tissue: return records associated with a particular tissue type. Currently, there are two tissue types: blood plasma and urine
- Publication: return records related to a particular published study

To the right of the page is the sample table, where users can browse the query results and perform the following actions:

- Click the checkbox to the left of each record to select/unselect, and then click the “Visualize” button to further inspect the selected records.
- Click the green plus sign to the right of each record to add them to the download list. Users can then click the “Download” button for the download page to download selected data files.

3.2. Visualization

Regarding the selected records, this page will display the fragment size distribution, as well as showing the coverage profile, fragment size profile, and the WPS track.

To the upper left is the reference genome. Users can switch between hg19 and hg38 and inspect corresponding fragment data.

Another useful feature is importing sessions from external sources. WashU Epigenome Browser (<http://epigenomegateway.wustl.edu/browser/>) has a feature enabling its users to save a session and share the contents with other users or to save it for future reference. Leveraging such features, FinaleDB can integrate active tracks from the external browser to the visualization page, allowing users to view cfDNA fragmentation pattern tracks and other external tracks side by side.

To do so, first go to WashU Epigenome Browser and load any tracks of interest, then click “Session” under the “Apps” tab, and download the session file. Next, return to the FinaleDB visualization page, and upload the session file either by clicking the “Import session” zone or drag-n-drop.

By default, the Epigenome Browser only displays up to 10 tracks (not including the refGene track). This is because displaying a large number of tracks may have negative impact on the browser’s performance. However, users can choose to ignore the warning message and display all the tracks at once.

3.3. Download

In the download center, users can make finer selections about which files to download, and launch the download process. Currently, FinaleDB provides the download access for the following data:

- Fragment data file (.tsv)
- Fragment coverage profile (bigWig)
- Fragment size profile (bigWig)
- Windowed Protection Score (bigWig)

Due to the nature of the data files, FinaleDB can’t on-the-fly bundle the selected files into a single compressed archive for download. Therefore, when users click the download button, all selected files will be downloaded separately. Some browsers may apply restrictions for this multi-file download behavior. In this case, please refer to the browser documentation for more information.

4. System design

The entire FinaleDB system consists of three parts:

1. Metadata database
2. Static file server
3. Web portal server

4.1. Metadata database

This database stores all metadata information. It is a PostgreSQL database and comes with the following tables:

1. seqrun: each record represents a “sequencing run” for a specific sample. Each seqrun record is identified by a unique ID starting with “EE”, such as EE86284.
2. sample: each record contains information of a particular sample
3. publication: each record represents a publication of a specific study project

The database is implemented with Amazon Relational Database Service (RDS), which is a service provided by Amazon Web Services (AWS).

4.2. Static file server

While the metadata information is stored in the PostgreSQL database described above, all fragmentation pattern analysis results themselves are stored as static files, in standard bioinformatics data formats.

To be more specific, these files include:

1. Alignments are stored as BAM files
2. Fragment files are stored as gzipped and indexed tab-separated value (TSV) files
3. Coverage, fragment profiles and WPS tracks are stored as bigWig files for genome browser visualization
4. QC reports (FastQC, Picard GC bias and library complexity estimation) are stored in plain text files

All the files are stored in an AWS S3 bucket which is publicly available over HTTPS.

4.3. Web portal

The web portal is a single-page app (SPA) developed on React.js (v16, <https://reactjs.org/>), a popular JavaScript web development framework, as well as client-side interface libraries listed below:

- Tabler-react (<http://tabler-react.com/>) for UI components
- Bootstrap React (<https://react-bootstrap.github.io/>) for UI components
- Highcharts (<https://www.highcharts.com/>) for interactive charts in the visualization page
- WashU Epigenome Browser for the displaying epigenomic tracks in the visualization page

The source code for the web portal is publicly available at https://github.com/epifluidlab/finaledb_portal under MIT License.

5. Programmatic access

In addition to the web interface, FinaledB provides a set of APIs for programmatic access. We believe it facilitates researchers to make the most of the database.

Below is a brief description of the API set.

5.1. Database summary

Endpoint: <http://finaledb.research.cchmc.org/api/v1/summary>

Usage: return summarized statistics about FinaledB database, such as the number of samples, a list of diseases present in the database, tissue types, etc.

5.2. Database queries

Endpoint: <http://finaledb.research.cchmc.org/api/v1/seqrun?limit=5>

Usage: return sequencing runs matching the query terms specified in the query string. For example, the API request below returns all records related to (Supplementary Table 4):

Query term	Description	Example
limit	limit the number of returned items	Limit the number of returned records by 2:

		http://finaledb.research.cchmc.org/api/v1/seqrun?limit=2
offset	specify the offset of the returned items list	Set the offset to 5: http://finaledb.research.cchmc.org/api/v1/seqrun?limit=2&offset=5
search	specify a search term, which can be either the seqrun ID, the SRA ID, or the sample name	Search by SRA ID: http://finaledb.research.cchmc.org/api/v1/seqrun?search=SRR2130049 Search by sample name: http://finaledb.research.cchmc.org/api/v1/seqrun?search=IH02
tissue	search for records related to a particular tissue type	Search for urine samples: http://finaledb.research.cchmc.org/api/v1/seqrun?tissue=urine
disease	search for records related to a particular pathological condition	Search for Hepatitis B samples: http://finaledb.research.cchmc.org/api/v1/seqrun?limit=5&disease=Hepatitis%20B
frag_num	format of the value should be <i>min,max</i> , where <i>min</i> and <i>max</i> defines a range for the fragment number of the returned records. For example, frag_num=10000000,100000000 returns records with the fragment number larger than 10M but less than 100M.	Return records with the fragment number larger than 10M and less than 100M: http://finaledb.research.cchmc.org/api/v1/seqrun?limit=5&frag_num=10000000,100000000
instrument	search for records generated by the specified sequencer	Search for records sequenced by Illumina HiSeq 2000: http://finaledb.research.cchmc.org/api/v1/seqrun?limit=5&instrument=HiSeq%202000

Supplementary Table 4. Function for RESTful API.